Yashika Gupta

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Education

Mahindra University Hyderabad, India
Bachelor of Technology in Computer Science Engineering — CGPA: 8.2/10 *Aug 2023 – Present*
Relevant Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Data Structures & Algorithms, Database Management Systems, Operating Systems, Theory of Computation, Object-Oriented Programming

Projects

Privacy Policy Grader — AI-Powered NLP Analysis Platform May 2026 – Present
GitHub

- Developed and deployed an AI-powered web application that evaluates privacy policies using custom NLP metrics, LLM-assisted analysis, and automated grading.
- Engineered an end-to-end pipeline for web scraping, readability scoring, legal-jargon detection, dark-pattern identification, and policy benchmarking.
- Implemented a hallucination-verification system that cross-validates LLM outputs against source documents to improve reliability and reduce unsupported claims.
- Built REST APIs using Flask and deployed the platform on Render with policy comparison, PDF export, caching, and interactive visual analytics.
- **Tech:** Python, Flask, Gemini API, BeautifulSoup, Selenium, SQLAlchemy, SQLite

Digital Stress Contagion in Social Network Jan 2026 – Apr 2026
GitHub

- Developed an agent-based simulation framework to model digital stress propagation in social networks using empirically derived smartphone usage behaviors.
- Applied K-Means clustering to construct behavioral personas and generated synthetic social networks using SBM and LFR network models.
- Modeled stress contagion through peer influence, resilience, and usage reinforcement dynamics while evaluating multiple intervention strategies.
- Demonstrated that hub-targeted interventions consistently outperformed random and peripheral-node approaches in reducing network-wide stress and burnout.
- **Tech:** Python, Scikit-learn, NetworkX, Agent-Based Modeling, K-Means, Data Analysis

Predicting Human Annotator Disagreement in CIFAR-10 Images Jan 2026 – Apr 2026
GitHub

- Adapted a ResNet-18 with 3 architectural modifications, applied two-phase training (50K hard label and 10K soft label). It is a deep learning model to predict full human label distributions rather than single-label classifications.
- Trained and evaluated models using KL Divergence, Jensen-Shannon Divergence, and entropy-based metrics to model annotator uncertainty.
- Analyzed relationships between model uncertainty and human disagreement through distributional evaluation and calibration studies.
- Performed explainability and robustness analysis using Grad-CAM visualizations and image corruption experiments.
- **Tech:** Python, PyTorch, ResNet-18, Grad-CAM

Skills

Programming Languages: Python, Java, C, JavaScript

Machine Learning & AI: PyTorch, Scikit-learn, XGBoost, NLP, Computer Vision, Deep Learning, Gemini API

Data Science: Pandas, NumPy, Matplotlib, Statistical Analysis

Backend & Databases: Flask, FastAPI, REST APIs, SQLAlchemy, PostgreSQL, MongoDB

Tools: Git, GitHub, Linux

Certifications

- Oracle Cloud Infrastructure (OCI) AI Foundations Certification (View Certificate)
- Oracle Cloud Infrastructure (OCI) Generative AI Professional Certification (View Certificate)
- Hackathon First Place in Web-Dev (View Hackathon Certificate)

Interests

Machine Learning, Artificial Intelligence, Computational Social Systems, Problem Solving, Basketball, Badminton